WEEK 8

Student Performance Dashboard

KAMALI V

BSC DATA SCIENCE & ARTIFICIAL INTELLIGENCE

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1. Introduction:

This report documents the Week 8 task, in which the data model prepared in Week 7 was turned into an interactive Student Performance Dashboard in Power BI Desktop. The dashboard lets a reader move from a high-level summary of academic performance to a second page of detail, and shows how background factors such as parental education, test preparation and lunch type relate to scores.

1.1 Aim:

To design, build, format and interpret a dashboard that answers key academic questions about student performance using the Math, Reading and Writing scores in the dataset.

1.2 Objectives:

- Verify the Week 7 data model and DAX measures before building visuals.
- Create KPI cards for the Math, Reading, Writing and Overall Average Score.
- Build visuals comparing subjects, parental education, test preparation, lunch type, pass/fail result and grade distribution.
- Create a second page named Academic Profile with detail visuals for the dataset.
- Add a Back button so the reader can return to the main dashboard.
- Format the dashboard with a consistent background and card style.

2. Dataset and Data Model Verification:

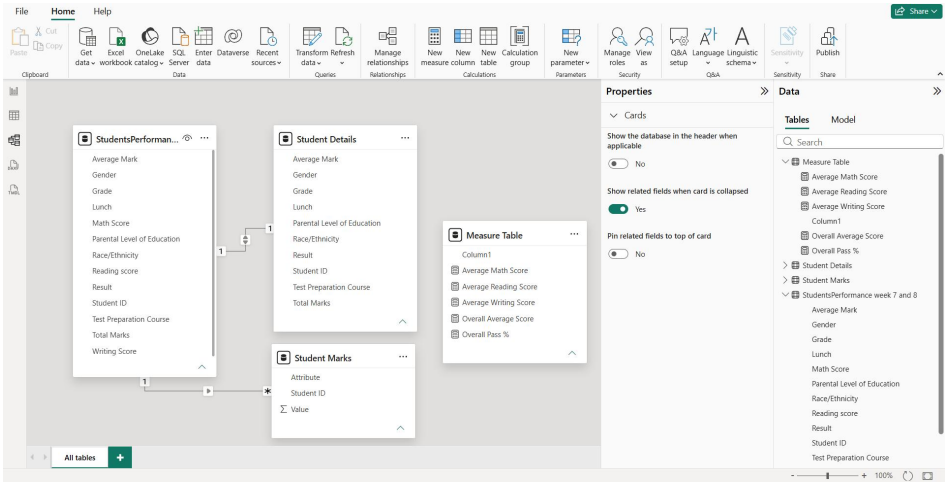
2.1 Tables in the Model:

Table	Key fields / measures
Student Details	Student ID, Gender, Grade, Lunch, Parental Level of Education, Race/Ethnicity, Result, Test Preparation Course, Total Marks, Average Mark
Student Marks	Student ID, Attribute (math, reading and writing score) and Value, the unpivoted form used for subject comparisons
StudentsPerformance (week 7 and 8)	Average Mark, Gender, Grade, Lunch, Math Score, Reading Score, Writing Score, Parental Level of Education, Race/Ethnicity, Result, Student ID, Test Preparation Course, Total Marks
Measure Table	Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score, Overall Pass %

2.2 Verification Checks:

- Confirmed in Model view that Student ID links the tables: StudentsPerformance relates to Student Details (one-to-one) and to Student Marks (one-to-many).
- Confirmed that the Measure Table holds the five measures used for the KPI cards and charts.

- Confirmed that the measures return plausible values, with subject averages between 0 and 100.
- Confirmed that the Grade, Result, Lunch, Parental Level of Education and Test Preparation Course fields are available for the visuals.



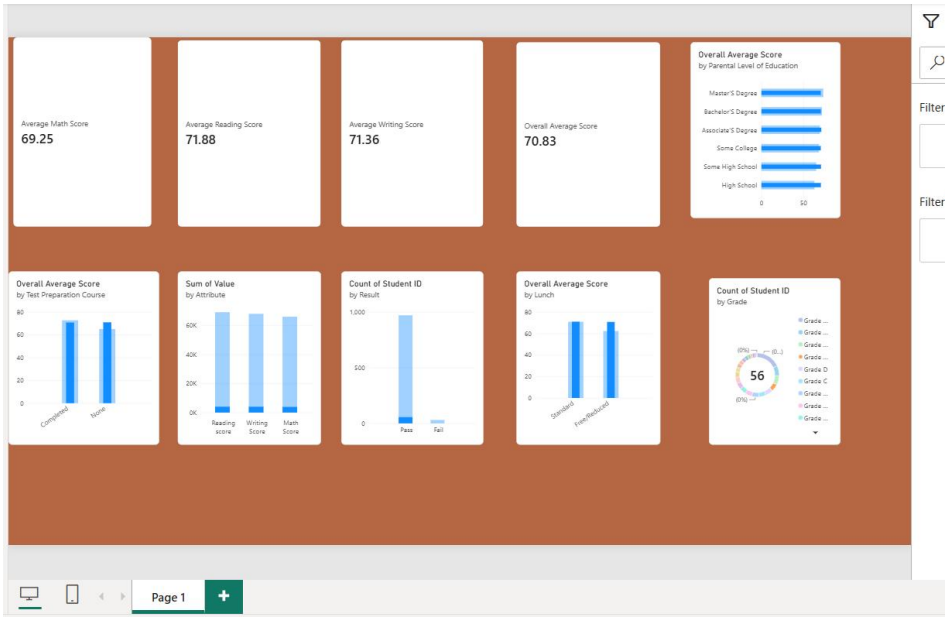
3. Dashboard Questions:

Dashboard question	Visual(s) that answer it
What is the overall academic performance?	KPI cards: Average Math, Reading, Writing and Overall Average Score
Which subject has the highest or lowest average?	Subject KPI cards; Sum of Value by Attribute column chart
Does test preparation relate to performance?	Overall Average Score by Test Preparation Course (column chart)
How does parental education relate to scores?	Overall Average Score by Parental Level of Education (bar chart)
How does lunch type relate to performance?	Overall Average Score by Lunch (column chart)
How many students passed or failed?	Count of Student ID by Result (column chart)
How are students spread across grades?	Count of Student ID by Grade (donut chart)

4. KPI Cards:

Four KPI cards were placed along the top of the dashboard so that the headline numbers are the first thing a reader sees. Each card is driven by a measure from the Measure Table.

Card	Field / measure	Value shown
Math Average	Average Math Score	69.25
Reading Average	Average Reading Score	71.88
Writing Average	Average Writing Score	71.36
Overall Average	Overall Average Score	70.83



5. Subject Performance and Parental Education Analysis:

5.1 Subject Performance:

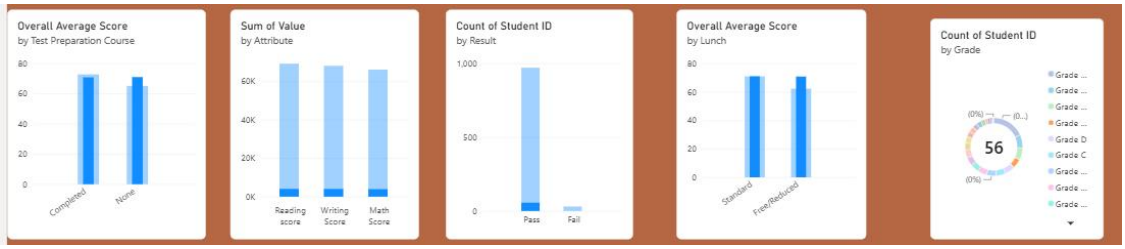
The three subject cards and the Sum of Value by Attribute column chart compare the subjects. The Attribute field from the unpivoted Student Marks table (math score, reading score, writing score) is used on the axis. Reading has the highest average at 71.88, followed by Writing at 71.36, and Math is the lowest at 69.25, a gap of about 2.6 points between Reading and Math.



5.2 Parental Education:

A bar chart titled Overall Average Score by Parental Level of Education lists the six education levels: master’s degree, bachelor’s degree, associate’s degree, some college, some high school and high school. Students whose parents hold a master’s or bachelor’s degree sit at the upper end of the range, while the high school group sits at the lower end. This is an association observed in the data and is not evidence that parental education causes the difference.

6. Test Preparation, Lunch Type, Result and Grade Distributionp:



6.1 Test Preparation:

The Overall Average Score by Test Preparation Course chart compares students who completed the course with those who did not (none). Students who completed test preparation have the higher overall average score.

6.2 Lunch Type:

The Overall Average Score by Lunch chart compares the two lunch types, standard and free/reduced. Students with the standard lunch have the higher average, which suggests a link between the socio-economic background reflected by lunch type and scores.

6.3 Result and Grade Distribution:

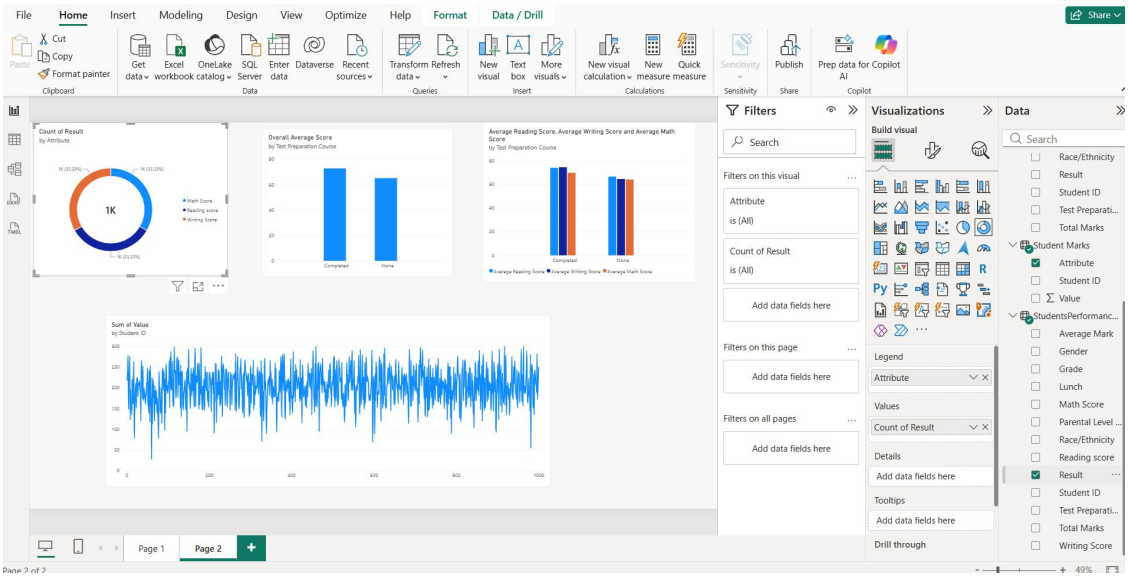
The Count of Student ID by Result chart shows that the large majority of students have a Pass result and only a small number Fail. The Count of Student ID by Grade donut chart shows how students are spread across the grade categories. The Grade field in this dataset is detailed, so the legend lists many categories and the donut has many small slices. The lighter bars in the charts show the full values and the darker overlay shows the portion highlighted by the current selection.

7. Academic Profile Page:

A second page, Academic Profile, was added to hold the detail behind the summary dashboard. It was built from the same data model, so its visuals use the same fields as the main page.

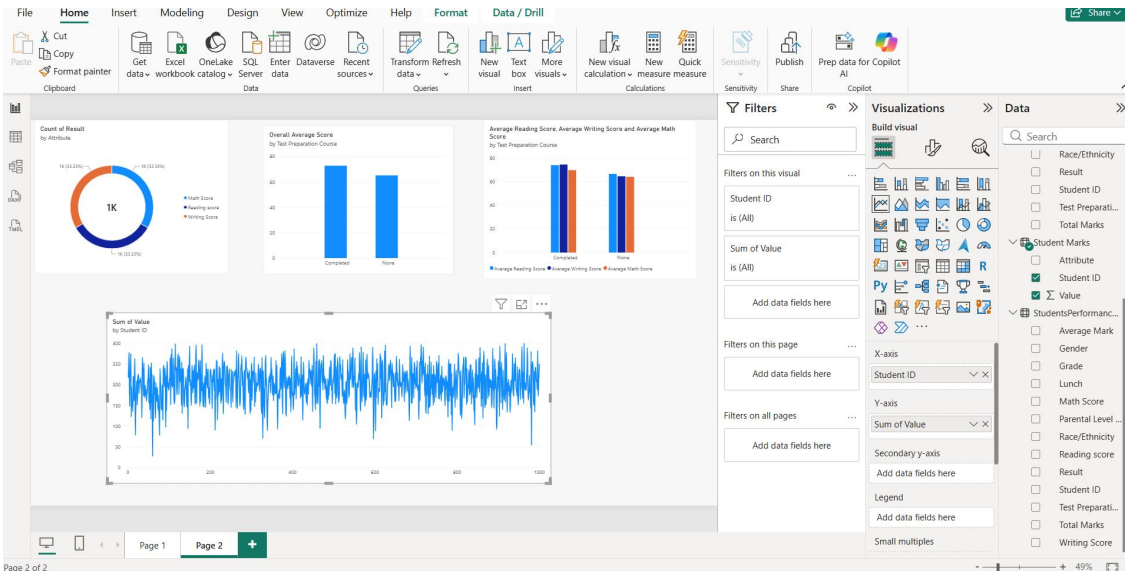
7.1 Result Count and Test Preparation Comparison:

The donut chart Count of Result by Attribute uses Attribute as the legend and Count of Result as the value. It shows a total of 1K, split equally (33.33% each) between math score, reading score and writing score. Beside it, two column charts show the Overall Average Score and the Average Reading, Writing and Math Score by Test Preparation Course, where students who completed the course score higher in all three subjects.



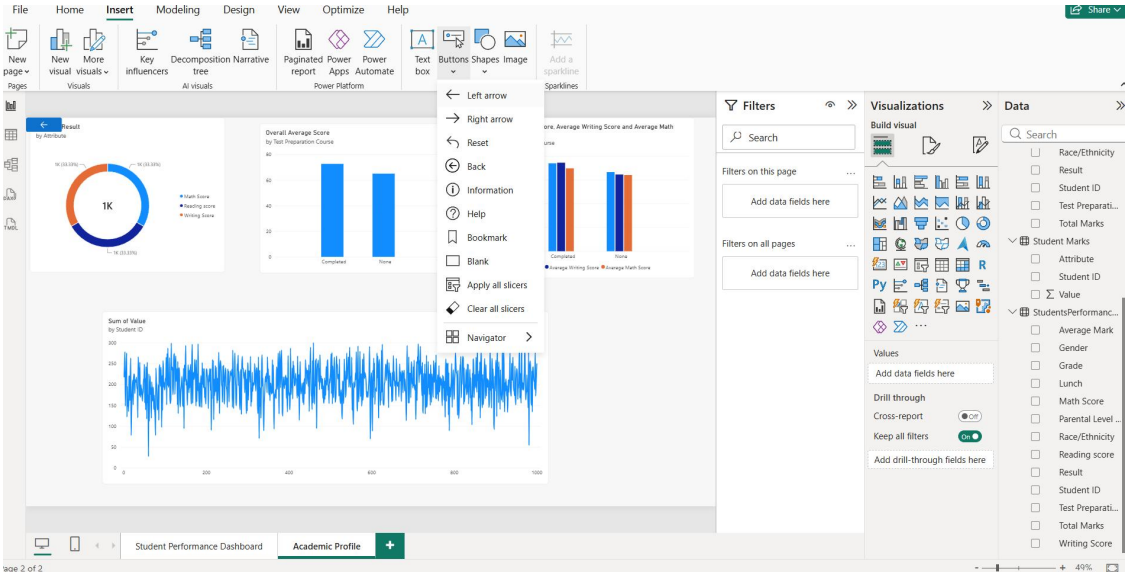
7.2 Total Score by Student ID:

A line chart was added with Student ID on the X-axis and Sum of Value on the Y-axis, so each point is the total of a student's three subject scores. Student IDs run from 0 to about 1000, and most totals fall between roughly 150 and 270 out of a possible 300, with a few sharp dips that mark students with very low scores.


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Format painter
Paste
Clipboard
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Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Prep data for AI
Copilot
Count of Result
by Attribute
1K
Math Score
Reading score
Writing Score
Overall Average Score
by Test Preparation Course
Completed
None
Average Reading Score, Average Writing Score and Average Math Score
by Test Preparation Course
Sum of Value
by Student ID
Filters
Search
Filters on this visual
Student ID
is (All)
Sum of Value
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
X-axis
Student ID
Y-axis
Sum of Value
Secondary y-axis
Add data fields here
Legend
Add data fields here
Small multiples
Data
Search
Race/Ethnicity
Result
Student ID
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Total Marks
Student Marks
Attribute
Student ID
Value
StudentsPerformanc...
Average Mark
Gender
Grade
Lunch
Math Score
Parental Level ...
Race/Ethnicity
Reading score
Result
Student ID
Test Preparati...
Total Marks
Writing Score
Page 1
Page 2
Page 2 of 2
49%

8. Back Button and Page Navigation:

The two pages were renamed Student Performance Dashboard and Academic Profile. A Back button was inserted on the Academic Profile page using Insert > Buttons > Back, and placed in the top-left corner of the page. When the reader clicks it, Power BI returns them to the page they came from, which is the main dashboard. The Buttons menu also offers options such as Left arrow, Right arrow, Reset, Bookmark and Navigator.



9. Dashboard Formatting and Theme:

Formatting was applied so that the dashboard is readable at a glance. A brown page background was used so that the white cards and charts stand out, and the same card style, title style and spacing were repeated across all visuals. The KPI cards sit in the top row, the comparison charts in the row below, and the Academic Profile page uses the same layout approach so that both pages look like one report (see Figure 2).

10. Conclusion:

The dashboard answers the questions defined at the start of the activity. The KPI cards give a quick view of overall performance, the subject and parental education charts show where scores differ, and the test preparation, lunch, result and grade visuals compare groups of students. The Academic Profile page adds detail by test preparation and by student, and the Back button returns the reader to the summary.

The main findings are that Reading is the highest-scoring subject (71.88) and Math the lowest (69.25), with an overall average of 70.83. Students who completed test preparation and students with the standard lunch have higher averages, and scores vary with parental education. Most students have a Pass result. These are associations in one dataset and should not be treated as proof of causes.